COURSE TITLE: CLOUD NATIVE DEVOPS			
Course Code:	**22PCA611**	**Course Type:**	**SPZ**
Teaching Hours/ Week(L:T:P):	**4:0:2**	**Credits:**	**5**
Total Teaching Hours:	**60+0+30**	**CIE + SEE Marks:**	**50+50=100**

Preamble *(brief description).*

In this course, we will explore the concepts, principles, and best practices related to cloud native development and DevOps methodologies. Whether you are a software developer, system administrator, or IT professional, this course will provide you with valuable insights and practical knowledge to excel in the world of cloud native DevOps.

Course Outcomes

At the end of the course students will be able to…

CO1	Implement Agile methodologies and utilize collaboration tools for software development.
CO2	Deploy Continuous Integration/Continuous Development pipelines in cloud environments
CO3	Describe about the collaboration and communication.
CO4	Discuss successful DevOps and CI/CD case studies for reference.
CO5	Use Infrastructure as Code (IaC) best practices for software delivery.

PO – CO mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	3	3	2	2	1	2
CO2	3	3	3	3	2	2	1	2
CO3	2	2	2	2	3	1	1	2
CO4	2	2	2	3	2	1	1	2
CO5	3	3	3	3	2	2	1	2

Note: '-'=Not Relevance; '1'=Low Relevance; '2' = Medium Relevance; '3'=High Relevance

Course content

Theory

Unit 1 Introduction to DevOps **10 Hours**

Service Pillars Overview; Operational Excellence, Security, Reliability, Performance Efficiency, Cost Optimization. Fundamental AWS Services; Setting up and Accessing your AWS Account, Cloud compute in AWS, Virtual Private Cloud Networking and Route Networking, Cloud Databases, Message and Queuing Systems, Trusted Advisor.

Unit 2 Implementation of a DevOps Culture **12 Hours**

Identity and Access Management and Working with Secrets and in AWS, Understanding shared responsibility Model in AWS, IAM roles, groups and policies, Integrating Federation with an AWS Account, Storing Secrets securely with AWS, Using Cognito with Application Authentication

Amazon S3 blob storage, S3 concepts, Endpoints, Access Control, Access Logs, Encryption options with S3, Using S3 events to trigger other AWS services, S3 Batch Operations, S3 replication and versioning.

Understanding the basics and background of DynamoDB, Data Modeling, Inserting and accessing Data, Streams DynamoDB accelerator, (DAX), Authenticating and Authorizing in DynamoDB and Monitoring DynamoDB

Unit 3 Continuous Integration and Delivery (CI / CD) using Infrastructure as Code (IaC)
18 Hours

Introduction to SDLC, Different Teams, Understanding different types of deployments, Essential CloudFormation topics, creating nested stacks with dependencies, Understanding how to detect drift in CloudFormation templates, using the Cloud Development Kit

Using CodeCommit for Code Versioning, Setup CodeCommit Repository, using AWS CodeBuild, About AWS Code Pipeline, setting up a code pipeline, using Jenkins to build your workload, About AWS CodeDeploy, use-cases of AWS CodeDeploy.

Understanding the Built-in Functionality of Elastic Beanstalk, Creating service role in IAM, Installing and using Elastic Beanstalk CLI, Deployment types in Beanstalk, Deploy an Application using Beanstalk

Lambda Deployment and Versioning, Lambda Functions, Lambda Triggers and event source mappings, Deploy versions using Lambda, Working with Lambda Layers, Monitoring Lambda functions.

Blue- Green Deployments, Understanding the Blue-Green Deployments, AWS Services that you can use for blue-green deployments, Benefits, Techniques for performing Blue-Green Deployments and Best Practices

Unit 4 Monitoring and Logging Environments and Workloads **10 Hours**

Cloud Watch and X-Ray's role in Devops, Overview, using cloudwatch to aggregate logs, cloudwatch alarms, adding application tracing with x-ray, Basic metrics in cloudwatch and creating dashboards, Amazon Event Bridge

Various Logs Generated; Power of AWS CloudTrail, Enabling Elastic Load Balancers, Using VPC flow logs

Unit 5
Enabling Highly Available Workloads and implementing Fault Tolerant workloads
10 hours

Understanding AWS Auto scaling, Deploying EC2 instance with Auto Scaling, Auto Scaling life Cycle, Using Auto Scaling life cycle hooks

Data Encryption Introduction, Understanding KMS Keys, Protecting data in transit with AWS certificate Manager,

AWS System Manager, AWS Config Essentials, Understanding Amazon Inspector, Configuring the inspector both manually and automatically

Detecting Threats with Amazon GuardDuty, Protect Data with Amazon Macie

Textbooks

1. Adam Book, "**AWS Certified DevOps Engineer – Professional** Certification and Beyond" Packt Publishing, February 2022

References

1. AWS official documentation of **AWS DevOps Engineer – Professional Certification** preparation material

Part B: Course Evaluation System

<table>
<tr><th>Assessment System</th><th>Assessment Component</th><th>Description</th><th>Weightage</th><th>Marks</th></tr>
<tr><td rowspan="6"><u>Theory</u>
Continuous Internal Evaluation (CIE)</td><td>CIE-I</td><td>Mid Semester Examination (MSE) –I of 1 hour duration for 25 marks</td><td>--</td><td rowspan="6">50</td></tr>
<tr><td>CIE-II</td><td>Mid Semester Examination (MSE) –II of 1 hour duration conducted for 25 marks</td><td>--</td></tr>
<tr><td>--</td><td>Average of MSE-1 and MSE-II</td><td>25</td></tr>
<tr><td>CIE-III</td><td>Assignments/Quizzes/Case Analysis</td><td>10</td></tr>
<tr><td>CIE-IV</td><td>Publications/Seminar/Quiz etc.</td><td>10</td></tr>
<tr><td>CIE-V</td><td>Attendance*</td><td>5</td></tr>
<tr><td colspan="3">CIE (Theory)</td><td>60%</td><td>*30M</td></tr>
<tr><td><u>Practical</u>
Continuous Internal Evaluation (CIE)</td><td>CIE-VI</td><td>Practical Test(s) of 30 Marks with Marks Distribution for each test as follows:
1; Write Up- 25 %
2: Execution – 35 %
3: Modification/Testing – 25%
4: Viva – 15 %
Continuous Evaluation in the form of Record Write Up, Viva etc ------ 20Marks</td><td>50</td><td></td></tr>
<tr><td colspan="3">CIE (Practical)</td><td>40%</td><td>*20 M</td></tr>
<tr><td>Semester End Evaluation (SEE)</td><td>SEE</td><td>Written Examination conducted for 3 hours duration **</td><td>50%</td><td>*50 M</td></tr>
<tr><td colspan="3">TOTAL MARKS</td><td>100%</td><td>100</td></tr>
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***Attendance Marks Allotment:** 90% and above : 5 Marks
85% to 89% : 4 Marks
80% to 84% : 3 Marks
75% to 79% : 2 Marks

****** Semester End Examination to be conducted for 3 hours duration for Courses with 3 or more credits; 2 hours duration for Courses with 2 credits and 1 hour duration for Courses with 1 credit.

Question Paper Pattern:

The SEE question paper comprises 5 Modules/Units. Each Module/Unit will have 2 questions carrying 20 marks each. Each question may have sub questions (a, b & c etc) for a total mark of 20. Students need to answer 5 questions selecting one FULL question from each module.

**50% weightage given to SEE and 50% weightage to CIE as per Scheme of evaluation